Juhwan Lee

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RESEARCH INTERESTS

Data-Driven Modeling and Decision Support for Proactive Airport and Air Traffic Operations.

- Airport Operations: Capacity analysis, demand-capacity balancing, and operational optimization
- Air Traffic Management: Aircraft trajectory analytics and flight operations performance assessment
- Passenger Terminal Operations: Passenger flow modeling, queue forecasting, and congestion management
- AI-Enabled Decision Support: Predictive modeling and intelligent operational decision-making

EDUCATION

Korea Aerospace University (KAU), Goyang, South Korea *Sep 2023 – Present*
Ph.D. in Air Transportation, Transportation and Logistics (Advisor: Dr. Hojong Baik)
Research focus: airport operations, capacity analysis, airport simulation, and data-driven forecasting

Korea Aerospace University (KAU), Goyang, South Korea *August 2023*
M.S. in Air Transportation, Transportation and Logistics (Advisor: Dr. Hojong Baik)
GPA: 4.0/4.0 (4.5/4.5)
Key courses: Introductory Applied Machine Learning, Airport Design with Simulation, Advanced Discrete Choice Theory in Transportation, Theory and Application of Data Mining, Airlines Operations Control

Korea Aerospace University (KAU), Goyang, South Korea *August 2021*
B.S. in Air Transportation, Transportation and Logistics
GPA: 3.77/4.0 (4.05/4.5)
Key courses: Air Traffic Management, Airspace System Analysis, Air Transportation System Analysis, Aviation Weather Service, Flight Operation Planning & Control, Computer Applications for Air Transportation Studies, Airport Operations

PROJECT EXPERIENCE

Integrated Information System Enhancement Project
(Sep 2025 - Mar 2026)

- Analyzed airport operational data, including flight schedules, passenger flows, queue lengths, and wait times.
- Developed and validated a CAST-based passenger terminal simulation model.

Strategic Planning for Enhancing Hub Competitiveness of Incheon International Airport
(Oct 2025 - Dec 2025)

- Analyzed changes in delay according to passenger demand and capacity conditions.
- Supported strategic evaluation of airport competitiveness from an operational performance perspective.

Development of Passenger Terminal Simulation Models for Incheon International Airport
(Jul 2025 - Dec 2025)

- Developed MATLAB-based simulations for departure processing facilities using actual terminal data.
- Estimated queue length and waiting time under time-varying demand and operational conditions.

Analysis of Variable Multiple Taxi-Line Operations for Apron Congestion Mitigation at Jeju International Airport
(Jul 2025 - Dec 2025)

- Simulated apron congestion and aircraft surface movement using TAAM.
- Evaluated taxi-line operational alternatives for congestion mitigation and surface movement efficiency.

Master Plan for Incheon International Airport
(Sep 2022 - Aug 2024)

- Performed runway capacity analysis using TAAM and passenger terminal simulation using CAST.
- Evaluated capacity, congestion, delay, and economic feasibility for airport master planning.

Performance-Based Navigation (PBN) Flight Procedure Maintenance Service
(Sep 2021 - Dec 2021)

- Analyzed observed aircraft trajectory data to evaluate the fuel efficiency of Continuous Descent Operations (CDO) and Continuous Climb Operations (CCO).

PUBLICATIONS

[1] **Juhwan Lee**, S. Yoon, K. Lee, S. Jeong, and H. Baik, "Airport Passenger Queue Forecasting for Departure Gates and Security Checkpoints," IEEE/AIAA 45th Digital Avionics Systems Conference (DASC), 2026.

[2] Y. Cho, J. Ahn, H. Baik, and **Juhwan Lee**, "Analysis of Aircraft De-icing/Anti-icing Efficiency Considering Pad Allocation and Ground Handling Operations," Journal of the Korean Society for Aviation and Aeronautics, 2026.

[3] J. Shin, H. Baik, **Juhwan Lee**, and B. Yi, "Network Connectivity Analysis of Major Asian Airports Using the NetScan Model:Focusing on Incheon Internation Airport," Journal of the Korean Society for Aviation and Aeronautics, 2025.

[4] J. Shin, B. Yi, M. Kim, J. Cho, and **Juhwan Lee**, "Study on Airport Capacity Utilization Analysis Using the Capacity Coverage Chart (CCC)," Journal of the Korean Society for Aviation and Aeronautics, 2025.

[5] E. Kim, **Juhwan Lee**, Y. Shin, and H. Baik, "A Study on Airport Access Mode Choice Behavior Using Mobile Data: Case of Incheon International Airport," IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC), 2023.

[6] C. Lee, **Juhwan Lee**, J. Park, and H. Baik, "An Analysis on the Utilization of STAR (Standard Terminal Arrival Route) and CDO (Continuous Descent Operation) Flight Ratio in the Domestic Airport," Journal of the Korean Society for Aviation and Aeronautics, 2022.

[7] Y. Lee, **Juhwan Lee**, and K. Lee, "A Study on the Visualization of an Airline's Fleet State Variation," Journal of the Korean Society for Aviation and Aeronautics, 2021.

AWARDS AND ACHIEVEMENTS

Best Paper of Session Award, AIAA DATC/IEEE Digital Avionics Systems Conference		*Fall 2026*
Scholarship, KAU Professional Talent Development Scholarship	$4,520	*Fall 2021*
2nd Place in Capstone Design Competition	$260	*Fall 2020*
2nd Place in ICT Startup Makerthon	$740	*Fall 2019*
Scholarship, Jeongseok C	$841	*Fall 2019*
Scholarship, Jeongseok C	$739	*Fall 2016*

TECHNICAL SKILLS

Programming	Python, MATLAB, LaTeX
Frameworks	PyTorch, NumPy
Simulation Tools	TAAM (Total Airspace and Airport Modeller), CAST (Passenger terminal simulation)